

1 Non-Directional Pre-Disclosure Drift Under Fair
2 Disclosure: Evidence from Korean Earnings
3 Announcements

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6 **Abstract**

Does South Korea's Fair Disclosure regulation effectively ensure that earnings information reaches all market participants simultaneously? Using 2,221 earnings disclosure filings from 64 KOSPI-listed companies across 12 chaebol groups (2020–2026), we find that pre-disclosure abnormal returns, while statistically significant under parametric tests ($CAR[-5, -1] = +0.52\%$, $p < 0.001$), do not predict the direction of actual earnings surprises (49.0% accuracy, $p = 0.649$; Spearman $\rho = -0.006$, $p = 0.876$). Non-parametric tests confirm that the drift is driven by outliers rather than systematic directional movement (sign test $p = 0.585$; Wilcoxon $p = 0.142$). In contrast, the first post-disclosure trading day (Day +1) responds sharply and correctly to earnings surprises: positive surprises generate AR of +0.55%, negative surprises -0.58% ($p < 0.001$), with 60.6% direction accuracy ($p < 0.001$). We further document that 96.5% of filings occur after market close, making Day 0 effectively pre-disclosure. Cross-sectional analyses reveal heterogeneous pre-disclosure drift across 12 chaebol groups (ranging from -0.23% to $+1.57\%$), but this variation is uncorrelated with event-window returns and inconsistent with differential information leakage. Firm size and market beta explain virtually none of the pre-disclosure drift ($R^2 = 0.002$). These results are robust to firm-clustered and month-clustered standard errors. Our findings suggest that Korea's Fair Disclosure regime effectively prevents directional information leakage, even though non-directional price noise and abnormal volume emerge in the pre-disclosure period.

7 *Keywords:* Fair disclosure, Earnings announcements, Pre-announcement drift,
8 DART, Korean stock market, Event study, Market efficiency

9 1. Introduction

10 Pre-announcement drift in stock prices is a well-documented phenomenon
11 across global equity markets (Kaniel et al., 2012). When stock prices move
12 before official earnings announcements, a natural concern arises: is material
13 information leaking to select market participants in violation of fair disclosure
14 regulations? This concern motivates securities regulators worldwide to mandate
15 simultaneous public dissemination of material information.

16 South Korea implemented its Fair Disclosure regulation in November 2002,
17 requiring listed companies to disclose material information simultaneously to all
18 market participants through the DART (Data Analysis, Retrieval and Transfer)
19 electronic filing system. While early studies found evidence of improved dis-
20 closure environments following the regulation’s adoption (Yun and Kim, 2012;
21 Oh and Park, 2017), whether the regulation continues to be effective over two
22 decades later remains an open question.

23 This study examines the long-term effectiveness of Korea’s Fair Disclosure
24 regulation by analyzing 2,221 earnings disclosure filings from 64 major KOSPI-
25 listed companies between 2020 and 2026. Our approach differs from prior
26 work in a critical respect: rather than simply documenting the existence of
27 pre-disclosure price movement, we test whether such movement is *directionally*
28 *informative*—that is, whether it predicts the sign of the actual earnings surprise.

29 Our findings reveal an important pattern. We observe statistically significant
30 pre-disclosure abnormal returns under standard parametric tests ($CAR[-5, -1]$
31 $= +0.52\%$, $p < 0.001$ under the market model; $p = 0.032$ with month-clustered
32 standard errors). However, this drift bears no systematic relationship to the
33 direction of actual earnings surprises. Pre-disclosure returns predict surprise
34 direction with only 49.0% accuracy—no better than a coin flip ($p = 0.649$). Non-
35 parametric tests (sign test, Wilcoxon signed-rank) fail to confirm the parametric

36 result, indicating that the observed mean drift is driven by outliers rather than
37 a pervasive pattern.

38 In contrast, the market responds sharply and correctly to the actual earnings
39 news on the disclosure date. Positive earnings surprises generate a $CAR[0,+1]$ of
40 $+0.81\%$, while negative surprises generate -0.47% , a highly significant difference
41 ($p < 0.001$). This finding is consistent with effective Fair Disclosure: material
42 information is incorporated into prices only when it becomes publicly available.

43 We make four contributions. First, we provide a recent assessment of Korea’s
44 Fair Disclosure regime using data spanning the COVID-19 and post-pandemic
45 periods (2020–2026). Second, we demonstrate the critical distinction between
46 the *existence* of pre-disclosure drift (which is ubiquitous) and its *directional in-*
47 *formativeness* (which would indicate information leakage). Third, we document
48 a previously overlooked institutional feature—96.5% of DART fair disclosure
49 filings are submitted after market close—that has important implications for
50 event study interpretation. Fourth, we provide detailed cross-sectional analyses
51 across chaebol groups, firm size terciles, and market beta terciles that reveal
52 significant heterogeneity in pre-disclosure drift but no pattern consistent with
53 differential information leakage.

54 The remainder of this paper is organized as follows. Section 2 reviews the
55 related literature. Section 3 describes our data and methodology. Section 4
56 presents the empirical results. Section 5 discusses the implications and limita-
57 tions, and Section 6 concludes.

58 2. Literature review

59 2.1. Fair disclosure regulation

60 Fair disclosure regulation addresses selective disclosure, whereby corporate
61 insiders share material information with favored market participants before pub-
62 lic release (Hefflin et al., 2003). In the U.S., Regulation Fair Disclosure (Reg FD),
63 implemented in 2000, produced mixed evidence: Ahmed and Schneible (2007)

64 find reduced information asymmetry, while Heflin et al. (2003) document re-
65 duced information quality. Bailey et al. (2003) provide comprehensive evidence
66 on market, analyst, and corporate responses to Reg FD, and Eleswarapu et al.
67 (2004) document reduced information asymmetry and trading costs following
68 implementation. Gomes et al. (2007) further show that Reg FD affected the
69 cost of capital, particularly for smaller firms.

70 The effectiveness of fair disclosure regulation depends critically on implemen-
71 tation and enforcement infrastructure. Christensen et al. (2016) demonstrate
72 that capital-market effects of securities regulation vary substantially with prior
73 institutional conditions and enforcement intensity. Jackson and Roe (2009) find
74 that regulatory intensity is positively associated with stock market development,
75 while Djankov et al. (2008) show that legal protections against self-dealing are
76 associated with more developed capital markets. Cumming et al. (2011) fur-
77 ther document that exchange trading rules, including disclosure requirements,
78 significantly affect market liquidity across 42 exchanges worldwide.

79 In Korea, Yun and Kim (2012) examined the effect of fair disclosure regula-
80 tion on the timeliness and informativeness of earnings announcements, while Oh
81 and Park (2017) examined how credit ratings interact with corporate disclosure
82 behavior under Korea’s fair disclosure regulation. Yoon et al. (2011) provide
83 early evidence on the impact of fair disclosure on the Korean stock market,
84 and Lee and Park (2012) test regulatory effectiveness using Korean data in an
85 emerging market context. Lee and Choi (2019) examine the interaction be-
86 tween R&D intensity and fair disclosure in Korea, finding that disclosure effects
87 vary by firm characteristics. However, these studies do not examine whether
88 pre-disclosure price movements are directionally informative with respect to the
89 actual earnings news—the central question of our study.

90 *2.2. Electronic disclosure systems*

91 A critical infrastructure underlying modern fair disclosure regulation is the
92 electronic filing system through which disclosures are disseminated. The U.S.
93 SEC’s EDGAR system, introduced in 1996, fundamentally changed how cor-

94 porate information reaches market participants (Bushee and Leuz, 2005). Ko-
95 rea’s DART system, operated by the Financial Supervisory Service since 2001,
96 serves an analogous function: all listed companies must file material disclosures
97 through DART before or simultaneously with any other dissemination channel.

98 The design of electronic disclosure systems has important implications for
99 information processing and market efficiency. DART provides structured fil-
100 ing templates for Fair Disclosure of Operating Results, enabling standardized
101 comparison across firms and time periods. Li (2010) surveys the literature on
102 textual analysis of corporate disclosures, demonstrating that format and read-
103 ability significantly affect how efficiently markets process information. Loughran
104 and McDonald (2011) show that the specific language used in financial filings
105 contains predictive information for future returns.

106 The timestamp precision of electronic filing systems is particularly relevant
107 to event study research. DART records receipt timestamps to the second, en-
108 abling researchers to classify filings as occurring during or after trading hours.
109 As we document, 96.5% of earnings disclosures are filed after market close, a
110 fact with fundamental implications for event study window interpretation that
111 prior studies have not fully exploited.

112 *2.3. Pre-announcement drift and its interpretation*

113 Pre-announcement drift in stock prices is well-documented but its interpre-
114 tation is debated. Meulbroek (1992) provides evidence of illegal insider trading
115 preceding announcements. Bernile et al. (2016) show that informed trading
116 occurs even ahead of macro-news announcements where information is tightly
117 controlled. However, Kaniel et al. (2012) demonstrate that individual investor
118 trading patterns around earnings announcements can generate drift without
119 requiring illegal information leakage.

120 Critically, the mere existence of pre-announcement drift does not establish
121 information leakage. Predictable earnings patterns allow sophisticated investors
122 to position ahead of announcements through legal means (Atiase, 1985). Chae
123 (2005) shows that abnormal volume before announcements reflects information

124 asymmetry but may arise from varying levels of investor sophistication rather
125 than illegal tip-offs. A related literature on post-earnings-announcement drift
126 demonstrates that markets may take days or weeks to fully incorporate earn-
127 ings information (Bernard and Thomas, 1989, 1990), further complicating the
128 interpretation of short-window event studies. Jegadeesh and Titman (2001) doc-
129 ument momentum effects that can generate pre-announcement drift patterns,
130 while Griffin et al. (2003) show that momentum profits persist across interna-
131 tional markets, including emerging markets like Korea.

132 The distinction between directional and non-directional drift is central to
133 our analysis. Non-directional drift—where prices move before announcements
134 but not in the direction of the forthcoming news—can arise from increased
135 attention (Hirshleifer and Teoh, 2003), disagreement among investors (Bamber
136 et al., 1997), or microstructure effects (Hasbrouck, 2003). Only *directional* drift
137 constitutes evidence consistent with information leakage.

138 2.4. *Event study methodology*

139 The standard event study framework (Brown and Warner, 1985; MacKinlay,
140 1997) relies on parametric tests of mean abnormal returns. However, Kolari
141 and Pynnonen (2010) demonstrate that cross-sectional correlation of abnormal
142 returns can substantially inflate test statistics, particularly when events cluster
143 in calendar time. Kothari and Warner (2007) provide a comprehensive review
144 of event study econometrics, emphasizing the importance of benchmark model
145 selection and cross-sectional dependence adjustments. We incorporate both non-
146 parametric alternatives and standardized returns to address these concerns.

147 2.5. *Korean market context*

148 The Korean stock market features concentrated chaebol ownership, a manda-
149 tory electronic disclosure system (DART), significant retail investor participa-
150 tion, and relatively strict insider trading regulations. Beny (2005) demonstrates
151 that stronger insider trading enforcement is associated with more efficient stock
152 prices across countries.

153 The chaebol structure is distinctive: large family-controlled conglomerates
154 such as Samsung, Hyundai Motor, SK, and LG account for a substantial share
155 of KOSPI market capitalization. Affiliates within the same chaebol group may
156 share information through internal channels, creating potential for correlated
157 trading patterns. At the same time, the diversity across chaebol groups provides
158 useful cross-sectional variation for testing whether disclosure regulation operates
159 uniformly across different types of firms. These institutional features, combined
160 with the comprehensive DART database, make Korea an ideal setting for testing
161 fair disclosure effectiveness.

162 **3. Data and methodology**

163 *3.1. Data sources*

164 We collect all Fair Disclosure of Operating Results filings from DART be-
165 tween January 2020 and March 2026 using the OpenDartReader API. We focus
166 on original filings, excluding corrections. Our sample comprises 2,221 disclosures
167 from 64 companies across 12 chaebol groups: Samsung (12 affiliates), Hyundai
168 Motor (8), LG (8), SK (7), HD Hyundai (5), Hanwha (5), POSCO (4), Lotte
169 (4), Doosan (4), CJ (3), KT (2), and GS (2). Daily stock price and volume data
170 are obtained from FinanceDataReader.

171 *3.2. Filing time classification*

172 We classify filing times using the DART receipt number prefix. In our sam-
173 ple, 96.5% of earnings disclosures carry the “800” prefix, indicating filing after
174 the 15:30 KST market close. This has fundamental implications: for these fil-
175 ings, the Day 0 return is entirely pre-disclosure, and the first trading day where
176 the market can react is Day +1.

177 *3.3. Market model*

178 We employ the market model with a $[-250, -30]$ estimation window (MacKin-
179 lay, 1997):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \epsilon_{i,t} \quad (1)$$

180 where $R_{m,t}$ is the KOSPI index return. We require a minimum of 60 trading
181 days in the estimation window, yielding 2,208 events with valid estimates. The
182 average estimated beta is 0.978. For robustness, we also report market-adjusted
183 returns ($\alpha = 0$, $\beta = 1$).

184 3.4. Earnings surprise measure

185 We extract year-over-year (YoY) operating profit changes from the disclosure
186 texts. Events where YoY operating profit increases are classified as positive
187 surprises; decreases as negative surprises. Of the 2,208 events with valid market
188 model estimates, 584 have parseable earnings surprise data: 358 positive and
189 226 negative surprises; unparseable cases primarily involve missing comparative
190 periods, newly listed firms, or format variations in disclosure texts.

191 We acknowledge that the standard measure in earnings announcement stud-
192 ies is the analyst consensus forecast surprise (Livnat and Mendenhall, 2006).
193 However, comprehensive analyst consensus data for Korean firms are not freely
194 available for our full sample period, and commercial databases require institu-
195 tional subscriptions. The YoY operating profit change, while crude, captures
196 the directional content of the earnings news and provides a conservative test.

197 3.5. Statistical tests

198 We report multiple test statistics to ensure robustness: parametric cross-
199 sectional t -tests, non-parametric tests (binomial sign test, Wilcoxon signed-
200 rank), and standardized abnormal returns. We also report firm-clustered and
201 month-clustered standard errors to address cross-sectional dependence (Kolari
202 and Pynnonen, 2010).

203 4. Empirical results

204 4.1. Pre-disclosure returns: Significant but non-directional

205 Table 1 presents daily abnormal returns. Parametric t -tests indicate sig-
206 nificant positive returns on several pre-disclosure days, particularly Day -1

(+0.256%, $t=4.57$). However, the sign test reveals that only 48–50% of individual returns are positive on each pre-disclosure day—no better than chance.

Table 1: Daily abnormal returns: Parametric vs. non-parametric tests

Day	AR (%)	Parametric		Sign Test		Standardized	
		t	p	% Pos	p	SAR	p
−5	0.101	2.09	0.037	50.5	0.609	0.040	0.048
−4	0.071	1.50	0.134	49.8	0.930	0.041	0.043
−3	0.086	1.71	0.088	49.3	0.562	0.022	0.279
−2	0.006	0.14	0.888	48.9	0.295	−0.002	0.935
−1	0.256	4.57	<0.001	50.1	0.909	0.089	<0.001
0	0.141	2.00	0.046	49.8	0.878	0.043	0.034
+1	−0.182	−2.39	0.017	48.3	0.109	−0.048	0.018
+2	0.023	0.43	0.669	49.2	0.541	0.022	0.275
+3	−0.118	−2.50	0.013	48.1	0.058	−0.043	0.033
+4	−0.047	−0.95	0.340	49.3	0.608	−0.006	0.765
+5	0.056	1.14	0.255	49.2	0.543	0.041	0.044

Note: Market model with $[-250, -30]$ estimation window ($N=2,208$). % Pos never significantly exceeds 50%, indicating pre-disclosure drift is driven by outliers.

Fig. 1 visualizes this pattern: while the cumulative AR line trends upward before disclosure, individual daily ARs show no consistent direction.

Table 2 confirms this pattern at the cumulative level. While the parametric $CAR[-5, -1]$ is +0.519% ($t=4.65$), the sign test shows only 49.4% of events have positive pre-disclosure CARs ($p=0.585$), and the Wilcoxon test is insignificant ($p=0.142$).

Notably, the $CAR[0, +1]$ window shows a significantly negative sign test result: only 47.0% of events have positive CARs ($p=0.003$). This asymmetry is consistent with the well-documented negativity bias in earnings reactions (Skinner, 1994).

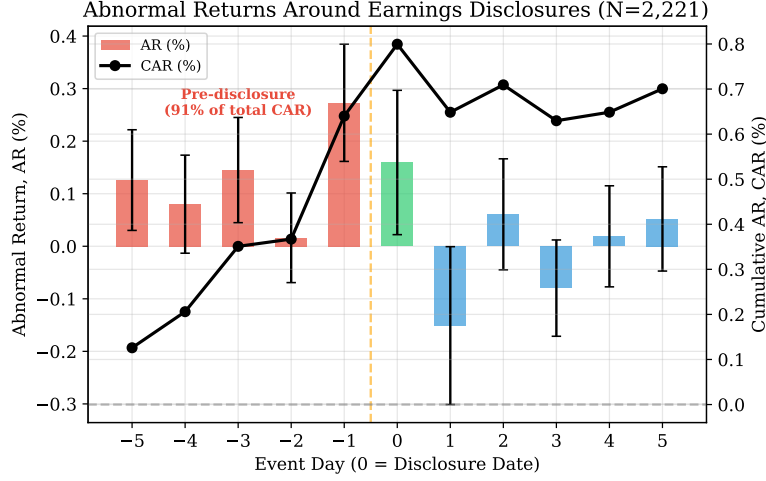


Figure 1: Daily abnormal returns (bars) and cumulative abnormal returns (line) around earnings disclosures ($N=2,208$). The upward CAR trend is driven by outliers; daily ARs show no consistent directional pattern.

219 4.2. The critical test: Does drift predict surprises?

220 Table 3 presents our central test. Pre-disclosure CARs show no difference
 221 by surprise direction ($CAR[-3, -1]$: $p=0.703$; $CAR[-5, -1]$: $p=0.192$). Direction
 222 prediction accuracy is 49.0% (binomial $p=0.649$), and Spearman ρ is -0.006
 223 ($p=0.876$).

224 In contrast, Day +1 returns respond sharply: positive surprises yield +0.55%,
 225 negative -0.58% ($p<0.001$), with 60.6% direction accuracy ($p<0.001$). This ex-
 226 tends to $CAR[0, +1]$ (+1.28%, $p<0.001$) and $CAR[+1, +5]$ (+0.96%, $p=0.010$).

227 With $N=584$, the test has 30.5% power to detect 53% true accuracy but
 228 99.8% power for 60%. We cannot rule out small-scale leakage, but any such
 229 effect would be economically modest.

230 4.3. Robustness

231 Table 4 confirms that results hold under both market-adjusted and market
 232 model specifications.

Table 2: Cumulative abnormal returns: Multi-test comparison

Window	CAR (%)	Parametric		Sign Test		Wilcoxon
		t	p	% Pos	p	p
$[-5, -1]$	0.519	4.65	<0.001	49.4	0.585	0.142
$[-3, -1]$	0.347	4.04	<0.001	48.1	0.060	0.946
$[0, +1]$	-0.042	-0.39	0.700	47.0	0.003	0.019
$[+1, +5]$	-0.268	-2.16	0.031	47.7	0.021	0.078
$[-5, +5]$	0.391	2.08	0.038	49.5	0.668	0.620

Table 3: CARs by earnings surprise direction ($N=584$)

Window	Positive ($N=358$)	Negative ($N=226$)	Diff.	p	Sig.
CAR $[-3, -1]$	+0.033%	+0.129%	-0.096%	0.703	
CAR $[-5, -1]$	+0.107%	+0.535%	-0.428%	0.192	
AR $[+1]$	+0.552%	-0.580%	+1.132%	<0.001	***
CAR $[0, +1]$	+0.813%	-0.470%	+1.283%	<0.001	***
CAR $[+1, +5]$	+0.481%	-0.481%	+0.962%	0.010	**

Note: Surprise defined by YoY operating profit change. Pre-disclosure CARs are indistinguishable by surprise direction; post-disclosure CARs respond correctly.

233 4.4. Abnormal volume

234 Volume increases from Day -3 ($1.05\times$, $p=0.004$), peaks on Day $+1$ ($1.71\times$),
235 and remains elevated through Day $+5$. Given non-directional returns, the pre-
236 disclosure volume increase likely reflects anticipatory positioning (Chae, 2005)
237 and disagreement among investors (Bamber et al., 1997; Berkman et al., 2009)
238 rather than informed directional trading.

239 4.5. Clustered standard errors

240 Table 5 reports t -statistics under alternative specifications. Under month-
241 clustered standard errors, CAR $[-5, -1]$ remains significant ($t=2.15$, $p=0.032$) but

Table 4: Robustness: Market-adjusted vs. market model

Window	Market-Adjusted ($N=2,221$)		Market Model ($N=2,208$)	
	CAR (%)	p	CAR (%)	p
$[-5, -1]$	0.640	<0.001	0.519	<0.001
$[-3, -1]$	0.434	<0.001	0.347	<0.001
$[0, +1]$	0.008	0.939	-0.042	0.700
$[+1, +5]$	-0.099	0.425	-0.268	0.031

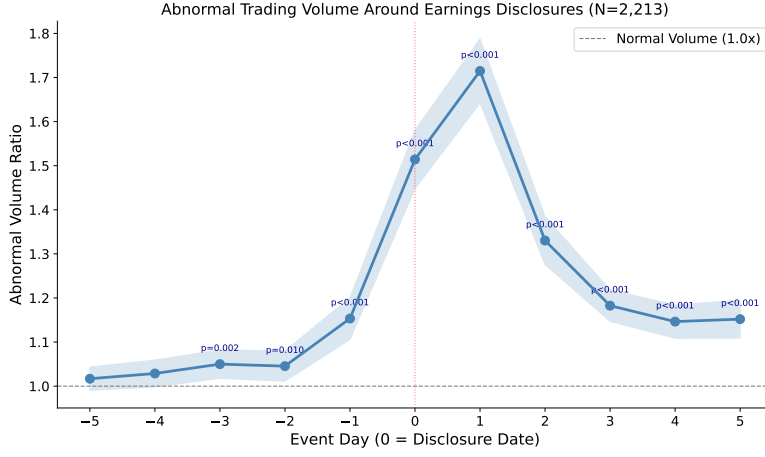


Figure 2: Abnormal trading volume around earnings disclosures ($N=2,213$). Volume peaks at $1.71\times$ normal on Day +1.

242 CAR $[-3, -1]$ becomes marginal ($t=1.80$, $p=0.072$), suggesting some parametric
 243 significance is inflated by event clustering.

244 4.6. Temporal and cross-sectional heterogeneity

245 We observe a structural shift in event-day CARs: positive during 2020–
 246 2022 (COVID-19 recovery) and negative during 2024–2026 (Fig. 3). A sig-
 247 nificant quarterly pattern emerges, with Q1 disclosures generating negative
 248 CARs (-0.51% , $p=0.010$) and Q2 generating positive CARs ($+0.60\%$, $p=0.003$).
 249 Cross-sectional OLS of CAR $[-3, -1]$ on $\log(\text{size})$ and beta yields $R^2 = 0.002$,

Table 5: Clustered standard errors

Window	Standard		Firm-clustered		Month-clustered	
	t	p	t	p	t	p
CAR[-5,-1]	4.38	<0.001	3.92	<0.001	2.15	0.032
CAR[-3,-1]	3.47	<0.001	2.95	0.003	1.80	0.072
CAR[0,+1]	-0.31	0.760	-0.25	0.802	-0.18	0.861

250 confirming that firm characteristics explain virtually none of the pre-disclosure
251 drift. Selection bias checks confirm parseable and non-parseable subsamples
252 show similar pre-disclosure CARs ($p=0.147$).

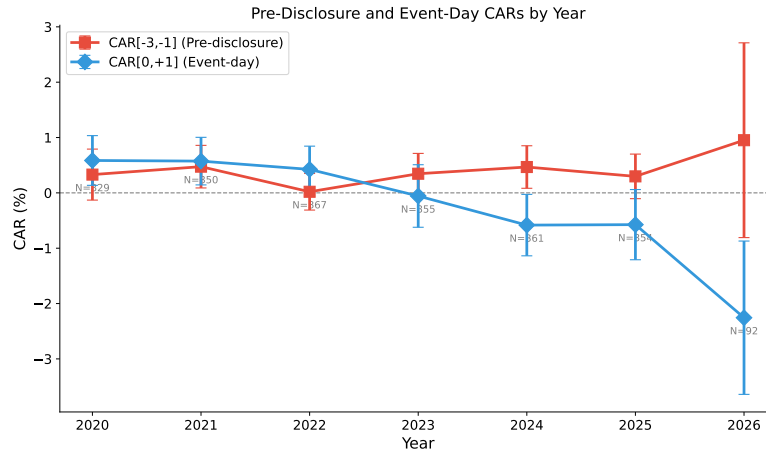


Figure 3: Pre-disclosure CAR[-3,-1] and event-day CAR[0,+1] by year. The temporal shift reflects macroeconomic conditions rather than changes in information leakage.

253 4.7. Cross-sectional analysis by chaebol group

254 The Korean market's chaebol structure provides a natural cross-sectional di-
255 mension for examining whether pre-disclosure drift varies systematically across
256 corporate groups. If information leakage were concentrated in specific chaebols—
257 perhaps due to weaker internal controls or more extensive analyst networks—we
258 would expect significantly higher directional drift in those groups.

Table 6 reports CAR[-3,-1] and CAR[0,+1] by chaebol group. Pre-disclosure CARs range from -0.228% (Doosan) to +1.565% (Hanwha), with Hanwha's drift significantly higher than the rest ($t=3.41$, $p<0.001$). An ANOVA test across all 12 groups yields $F=1.73$ ($p=0.062$), suggesting marginally significant heterogeneity.

Table 6: Cumulative abnormal returns by chaebol group

Chaebol	N	CAR[-3,-1]		CAR[0,+1]	
		Mean (%)	t	Mean (%)	t
Hanwha	116	+1.565	2.30	-0.583	-0.81
CJ	75	+0.881	1.98	-0.120	-0.16
GS	53	+0.604	1.37	+0.674	1.53
Lotte	92	+0.517	1.33	+0.202	0.36
HD Hyundai	497	+0.474	2.38	+0.312	1.42
Samsung	292	+0.318	1.70	+0.012	0.05
LG	185	+0.278	1.22	-0.311	-0.81
SK	195	+0.127	0.54	-0.286	-0.84
Hyundai Motor	402	+0.093	0.52	-0.352	-1.62
POSCO	156	+0.060	0.15	-0.427	-0.86
KT	80	-0.006	-0.03	+1.310	4.25
Doosan	65	-0.228	-0.46	-0.677	-0.96
All	2,208	+0.347	4.04	-0.042	-0.39

Note: Sorted by descending CAR[-3,-1]. ANOVA across groups: $F=1.73$ ($p=0.062$) for CAR[-3,-1], $F=1.29$ ($p=0.223$) for CAR[0,+1].

Fig. 4 visualizes the cross-sectional pattern. Hanwha's elevated drift likely reflects the defense sector's sensitivity to geopolitical news rather than information leakage about earnings specifically. KT exhibits remarkably high event-window returns (+1.310%, $t=4.25$, $p=0.027$ vs. non-KT), consistent with telecommunications sector earnings surprises being particularly impactful. Samsung—

269 despite being the most closely watched chaebol—shows only moderate pre-
270 disclosure drift (+0.318%).

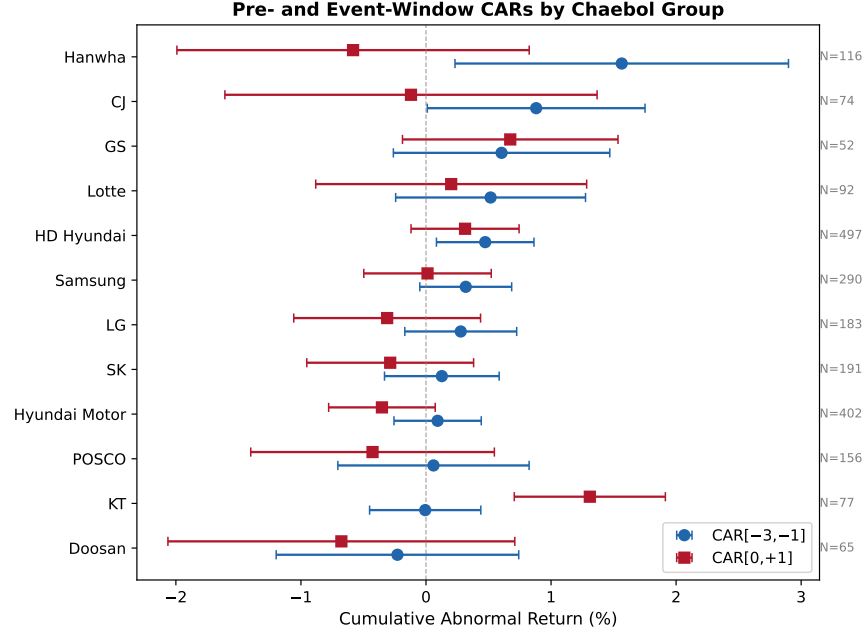


Figure 4: Pre-disclosure CAR[-3,-1] and event-window CAR[0,+1] by chaebol group with 95% confidence intervals. Substantial cross-group variation exists, but no group shows patterns consistent with systematic directional information leakage.

271 Importantly, the cross-group variation does not exhibit the pattern expected
272 under systematic information leakage. Pre-disclosure and event-window CARs
273 are uncorrelated across groups (Spearman $\rho = -0.14$, $p = 0.668$), suggesting that
274 the drivers of pre-disclosure drift are distinct from the earnings information
275 revealed at disclosure.

276 4.8. Firm size and beta effects

277 To investigate whether pre-disclosure drift varies with firm characteristics,
278 we divide the sample into terciles by firm size (log market capitalization) and
279 market beta. Table 7 reports CARs for each tercile.

280 Panel A reveals a monotonically decreasing relationship between firm size

Table 7: CARs by firm size and market beta terciles

Tercile	N	CAR[-3,-1]		CAR[0,+1]		CAR[-5,-1]
		Mean (%)	t	Mean (%)	t	Mean (%)
<i>Panel A: Firm size terciles</i>						
Small	790	+0.403	2.91	-0.058	-0.32	+0.578
Medium	726	+0.374	2.45	-0.005	-0.03	+0.590
Large	679	+0.250	1.58	-0.109	-0.60	+0.359
<i>Panel B: Market beta terciles</i>						
Low β	727	+0.515	3.53	-0.063	-0.35	+0.607
Mid β	732	+0.303	2.04	-0.033	-0.17	+0.446
High β	736	+0.222	1.45	-0.073	-0.38	+0.492

Note: Size tercile boundaries: Small ($\log \text{size} \leq 12.56$), Medium (12.56–13.45), Large (> 13.64). Beta tercile boundaries: Low ($\beta \leq 0.845$), Mid (0.845–1.129), High ($\beta > 1.130$).

and pre-disclosure drift. However, the small-large difference is not statistically significant ($t=0.92$, $p=0.358$), and a Kruskal-Wallis test is also insignificant ($H=2.88$, $p=0.237$). Panel B shows a similar pattern for market beta. The Spearman correlation between beta and CAR[-3,-1] is -0.057 ($p=0.008$), indicating a statistically significant but economically modest negative relationship. Low-beta stocks may exhibit higher pre-disclosure drift because their lower systematic risk makes firm-specific information relatively more important.

Event-window CARs (CAR[0,+1]) are uniformly close to zero across all size and beta terciles, confirming that the non-directional character of post-disclosure returns is not an artifact of pooling heterogeneous subsamples. These results are consistent with the well-documented size effect in pre-announcement drift (Atiase, 1985): smaller firms have less pre-disclosure information production, creating larger price adjustments when new information arrives.

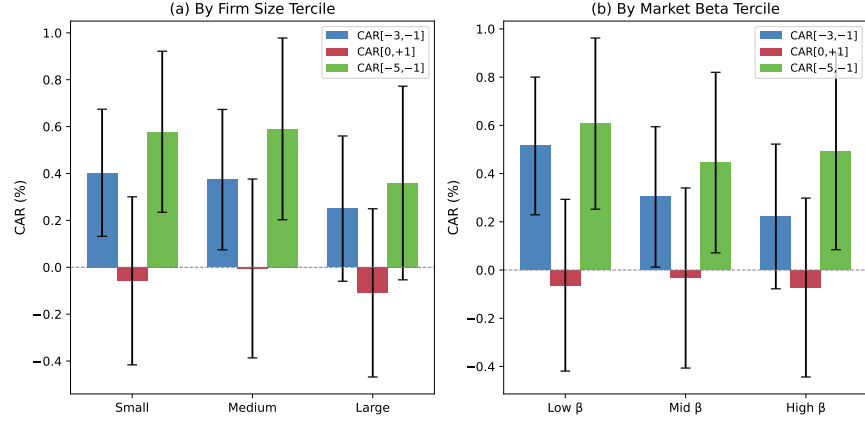


Figure 5: CARs by firm size tercile (Panel a) and market beta tercile (Panel b) with 95% confidence intervals. Pre-disclosure drift is slightly larger for smaller and lower-beta firms, but event-window CARs are uniformly near zero across all terciles.

5. Discussion

5.1. Drift without directional leakage

Three complementary tests suggest the drift is non-directional: (1) direction test: 49.0% accuracy; (2) non-parametric tests: sign test ($p=0.585$) and Wilcoxon ($p=0.142$) fail; (3) surprise conditioning: $p=0.703$. The drift appears driven by company-specific news unrelated to earnings, market momentum, or anticipatory trading based on publicly available signals. This interpretation is consistent with Hirshleifer and Teoh (2003), who argue that limited investor attention can produce predictable patterns without implying illegal information access.

The cross-sectional analyses reinforce this conclusion. While pre-disclosure drift varies across chaebol groups (Hanwha showing the highest at +1.565%), this variation is uncorrelated with event-window returns, inconsistent with directional information leakage. The size and beta patterns—smaller and lower-beta firms showing modestly larger drift—reflect normal variation in information environments rather than differential insider access. The most parsimonious explanation combines positive market momentum during the sample period,

311 right-skewed return distributions (Flannery and Protopapadakis, 2004), and
312 anticipatory non-directional trading (Chae, 2005).

313 5.2. *Comparison with international evidence*

314 Our findings align with the broader evidence on fair disclosure regulation.
315 Bailey et al. (2003) find that Reg FD reduced selective disclosure, and Eleswarapu
316 et al. (2004) document reduced information asymmetry. Christensen et al.
317 (2016) show that capital-market effects of securities regulation depend on prior
318 institutional conditions and enforcement intensity. Our contribution is distinct:
319 rather than evaluating the *introduction* of fair disclosure, we test whether the
320 regulation continues to be effective *within* the regulated regime by examining
321 directional content over a six-year window.

322 The Korean result contrasts with evidence from markets with weaker en-
323 forcement. Beny (2005) demonstrates that countries with stronger insider trad-
324 ing enforcement exhibit more efficient stock prices, and Leuz et al. (2003) find
325 that investor protection is associated with lower earnings management. Korea’s
326 combination of mandatory electronic filing, strict enforcement by the Finan-
327 cial Supervisory Service, and concentrated analyst coverage appears to create
328 an institutional environment where directional information leakage is effectively
329 deterred.

330 5.3. *Methodological contributions*

331 This study makes several methodological contributions. First, we demon-
332 strate the importance of **multi-test triangulation** in event studies. The para-
333 metric *t*-test implicitly assumes normally distributed abnormal returns. When
334 the distribution is right-skewed, the parametric mean is inflated by outliers, pro-
335 ducing significant *t*-statistics even when the majority of events show no drift.
336 The sign test and Wilcoxon signed-rank test provide a more accurate charac-
337 terization of the typical event.

338 Second, we introduce the **directional conditioning test** as a direct diag-
339 nostic for information leakage. Prior event studies typically interpret signifi-
340 cant pre-announcement drift as evidence of informed trading without verifying

341 whether the drift predicts the direction of the subsequent news. This test could
342 be adopted as a standard diagnostic in future event studies.

343 Third, we document the importance of **filing time classification**. The
344 finding that 96.5% of DART filings occur after market close implies that Day 0
345 returns are effectively pre-disclosure. Studies that attribute Day 0 returns to
346 information release may be misinterpreting anticipatory trading as immediate
347 market reaction.

348 *5.4. Implications for policy and practice*

349 For regulators, non-directional drift suggests Korea’s Fair Disclosure regime
350 does not require fundamental reform. Material earnings information reaches
351 the market simultaneously, and prices respond correctly only upon public dis-
352 closure. For investors, post-disclosure Day +1 returns correctly price surprises
353 with 60.6% accuracy ($p < 0.001$), while pre-disclosure drift strategies are unlikely
354 to be profitable directionally.

355 *5.5. Limitations*

356 Several limitations warrant discussion. First, our YoY operating profit
357 change as a surprise measure is crude compared to the analyst consensus forecast
358 standard (Livnat and Mendenhall, 2006). Only 26.4% of events yield parseable
359 surprises, and parseable events come from larger firms with lower beta. Sec-
360 ond, our sample is limited to 64 large chaebol affiliates; information leakage
361 may be more prevalent among smaller, less scrutinized firms. Third, Korean
362 Fama-French factors are unavailable, preventing multi-factor benchmark mod-
363 els. Fourth, we cannot rule out that pre-disclosure volume increases reflect legal
364 mosaic information assembly. Fifth, our analysis focuses on earnings disclosures
365 and does not examine other material events where leakage incentives may differ.

366 **6. Conclusion**

367 This study finds no evidence of directional information leakage in Korean
368 earnings announcements. Three independent analyses converge: direction pre-

369 diction accuracy of 49.0% (no better than chance), failure of non-parametric
 370 tests to confirm parametric drift (sign test $p=0.585$; Wilcoxon $p=0.142$), and
 371 surprise-conditioned CARs that are indistinguishable by earnings direction ($p=0.703$).
 372 The first post-disclosure trading day shows a strong, correct directional response
 373 (60.6% accuracy, $p<0.001$), confirming that material earnings information is in-
 374 corporated into prices only upon public release through the DART system.

375 Cross-sectional analyses reveal significant heterogeneity across chaebol groups,
 376 firm sizes, and market betas, but these patterns are consistent with normal
 377 variation in information environments rather than differential information leak-
 378 age. The most-watched chaebol group (Samsung) shows only moderate, non-
 379 directional drift.

380 Our results carry a broader methodological lesson: pre-announcement drift
 381 should not be conflated with information leakage without conditioning on the
 382 direction of the actual news (Meulbroek, 1992; Bernile et al., 2016; Atiase, 1985).
 383 Future work should extend the analysis to non-chaebol firms and incorporate
 384 analyst consensus surprises.

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 388 through DART (<https://dart.fss.or.kr>). Stock price data are from KRX
 389 via FinanceDataReader. Analysis code is available at [Availableuponpublication].

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